

Yunji Seo

Cambridge, MA | (857) 763-6094 | yunjiseo@mit.edu | [Personal Website](#) | [LinkedIn](#) | [GitHub](#)

RESEARCH INTERESTS

Mobility Equity & Accessibility | Urban Robotics | Human-Centered Autonomous Systems | Urban Mobility Systems

EDUCATION

Massachusetts Institute of Technology (MIT), MA Sep 2026 – May 2027 (Expected)

MEng in Civil and Environmental Engineering (Data Science for Engineering Systems)

Yonsei University, Seoul, South Korea Mar 2021 – Aug 2026

Bachelor of Science in Urban Planning and Engineering GPA: 4.21 / 4.3

- Relevant Coursework: Urban Engineering Capstone Design, Mobility Data Mining, Mobility Analysis and Planning, Smart Urban Transit, Transportation Planning, Smart Urban Design Software Tools, Residential Areas Planning and Studio, National and Regional Planning, Probability and Statistics for Urban Planning, Urban Spatial Information Analysis, R and Python Programming, Urban Planning, Environmental Planning, Urban Form and History, Fundamentals of Urban and Regional Economics, Land and Housing Policy, The Theory of Urban Design, Green City, Engineering Mathematics I-III

University of California, Berkeley, CA Jan 2025 – May 2025

Exchange student in Urban Studies GPA: 3.87 / 4.0

- Relevant Coursework: Introduction to Advanced Programming in R, Data Structures, Urban and Community Health, Community Planning and Public Policy for Disability

RESEARCH EXPERIENCE

Research Collaborator May 2026 – Present

JTL Urban Mobility Lab, Massachusetts Institute of Technology, Cambridge, MA

- Examining long-term public transit ridership trends across U.S. metropolitan areas using FTA NTD records.
- Integrating ridership data with socioeconomic, employment, and transportation infrastructure variables to support metropolitan-level panel and time-series analyses.

Visiting Student May 2026 – Aug 2026

Senseable City Lab Seoul, Massachusetts Institute of Technology, Seoul, South Korea

- Analyzed large-scale GPS mobility trajectory data to examine age-related segregation in individual mobility patterns and developing data cleaning and quality assessment procedures.

Student Research Assistant Nov 2025 – Feb 2026

Transportation Planning Lab, Yonsei University, Seoul, South Korea

- Processed smart card data to support a fare elasticity and pricing strategy study for the Shinbundang Line.

Student Research Assistant Apr 2025 – Aug 2025

California Partners for Advanced Transportation Technology (PATH), University of California, Berkeley, CA

- Contributed to Project 3: Personal Mobility on I-680 Corridor Demonstration under the CCTA Automated Driving Systems (ADS) program, focusing on evaluating automated vehicle performance in mixed traffic conditions.
- Analyzed vehicle trajectory and GPS data to assess safety performance metrics such as Time-to-Collision (TTC).
- Supported data processing pipelines using Python and assisted in designing visualizations to identify lane change patterns.

Urban Regeneration Company Internship Jun 2024 – Aug 2024

Renovaring Korea, Seoul, South Korea

- Assisted with a Ministry of the Interior and Safety-supported rural revitalization project, through program design, field engagement, and interviews to develop strategies for attracting retirees to an aging village in Jeju Island.
- Supported a local government project (Seodaemun District Office in Seoul) by assisting with the development and operation of an entrepreneurship education program aimed at revitalizing the Sinchon and Ewha neighborhoods affected by post-COVID economic decline.

SKILLS

Technical Skills

- **Programming:** Python, Java, R, Google Earth Engine
- **Design Tools:** QGIS, AutoCAD, Illustrator, Photoshop, SketchUp, Rhino 3D, InDesign, Adobe Premiere Pro

PROJECTS

Undergraduate Team Projects *Yonsei University, Seoul, South Korea*

Development of Accessibility Evaluation Indicators for Elderly Target Facilities and Analysis of DRT Suitability Using IPA Matrix: Focused on Northern Gyeonggi Province in a Super-Aged Society Fall 2024

- Identified the lack of accessibility evaluation indicators tailored for the elderly and developed a specialized framework. Analyzed areas with insufficient accessibility to medical facilities for the elderly in Northern Gyeonggi Province, highlighting regions requiring DRT services.
- Utilized Python for dynamic web crawling, IPA matrix for prioritization, and QGIS for spatial analysis. Created presentation panels using Photoshop and Illustrator.

Proposal for Mobility Solutions on Yonsei Campus Fall 2023

- Identified campus mobility challenges, proposed a paid mobility service, and conducted a stated preference survey using SAS orthogonal design. Estimated parameters using R's logit model analysis.

Evaluation of Public Housing Policy for the Elderly in Korea Fall 2023

- Investigated urban issues in Seoul, focusing on housing policies for the elderly and their mobility rights; produced a video outlining findings and solutions.

Residential Complex Design for Yongsan, Seobinggo Apartment Site Fall 2023

- Conducted site analysis and resident interviews, developed residential design concepts, and created 3D models and presentation panels using AutoCAD, Photoshop, Illustrator, and Rhino.

Revitalization Plan for Underutilized Spaces on Yonsei Sinchon Campus Fall 2023

- Observed campus areas, analyzed pedestrian patterns with Excel, and proposed activation strategies supported by 3D modeling.

Shuttle Bus Route Proposal for Yonsei Sinchon Campus Spring 2023

- Designed new routes based on survey data and performed B/C analysis to optimize shuttle bus operations.

Environmental Impact Assessment and Solutions for Abandoned Mines Spring 2023

- Conducted a case study on pollution caused by abandoned mines in Korea and proposed mitigation strategies.

New Night Bus Route Proposal Using Taxi O/D Data in Seoul Fall 2022

- Conducted spatial analysis using QGIS, R, and Excel to identify underserved areas and propose new bus routes for late-night transportation.

Analysis and Activation Proposal for Underutilized Areas on Yonsei International Campus Fall 2022

- Identified underutilized areas, proposed activation strategies, and created presentation panels using AutoCAD, Photoshop, and Illustrator.

HONORS AND AWARDS

Korea National Science & Technology Scholarship Fall 2024

Full tuition merit scholarship for two years

Korea-U.S. Advanced Field Youth Exchange Support Program Scholarship Fall 2024

Awarded KIAT's Future Mobility STEM scholarship (\$9,000) for a semester-long study abroad program

Korean Society of Transportation Award Fall 2024

Awarded for academic excellence in a senior capstone project on mobility for older adults

KAIT (Korea Association for ICT Promotion) Award Fall 2024

National data analysis competition (Participation in the Origin-Destination Data Analysis Category)

Highest Honors Fall 2022, Spring 2023, Fall 2024

Yonsei University, Department of Urban Planning and Engineering, Seoul, South Korea